Source and selective outcomes

Source image:
tomato leaf late blight
GT bounding box



1 Supported grounding

Query: Locate the tomato leaf late blight in this image.

Output: Supported grounding

Evidence: query–image consistent



2 Visual contradiction

Query: Where is the tomato early blight leaf in this image?

Output: Abstain — visual contradiction

Reason: visual contradiction (same image, incompatible query)



3 Target unavailable

Query: Locate the tomato leaf late blight in this image.

Output: Abstain — target unavailable

Reason: target unavailable (absent or completely excluded)



Seven deterministic relations

T1 Query paraphrase

Same image, target, and GT box



Answer preserved

T2 Photometric variation

Brightness = 1.1; contrast = 1.1



Answer preserved

T3 Geometric transformation

Original



Horizontally flipped



Answer preserved | GT box transformed

T4 Target-region degradation

(Dose-only; no hard abstention label)
Target-inclusion crop → degradation

Gaussian blur



Pixelation



Mean-color replacement



T4 control



Non-target background degraded
Target remains supported

T5 Absent cross-crop query

Query: Locate the apple rust leaf in this image.

Different crop



Output: Abstain— target unavailable

T6 Confusable same-crop query

Query: Where is the tomato early blight leaf in the image?

Same crop;confusable disease



Output: Abstain— visual contradiction

T7 Visibility control

T7 inclusion crop

Supported target-inclusion crop



T7 ladder

100%



75%



50%



25%



0%



Supported in construction | Dose-only | Target unavailable

Same-source counterfactual groups

Appearance or transformation identity alone does not determine the evidence state.

Base (source)



Base crop-out
0% visibility crop



Target unavailable

T2



T2 crop-out
0% visibility crop



Target unavailable

T3



T3 crop-out
0% visibility crop



Target unavailable

Contrasting outcomes under shared provenance make source identity alone insufficient for prediction.

A Supported grounding (query–image consistent)
C Visual contradiction (incompatible query)
U Target unavailable (absent or completely excluded)
Annotated target region
Degraded / removed region